

CStats - Lab #6

March 31, 2026

Exercise 1.

An engineer is interested in the effects of cutting speed (A), tool geometry (B), and cutting angle (C) on the life (in hours) of a machine tool. Two levels of each factor are chosen, and three replicates of a 2^3 factorial design are run. The results are as in Table 1.

- (a) Estimate the factor effects. Which effects appear to be large?
- (b) Use the analysis of variance to confirm your conclusions for part (a).
- (c) Write down a regression model for predicting tool life (in hours) based on the results of this experiment.
- (d) Analyze the residuals. Are there any obvious problems?
- (e) On the basis of an analysis of main effect and interaction, what factor levels of A, B, and C would you recommend using?

A	B	C	Treatment Combination	Replicate		
				I	II	III
–	–	–	(1)	22	31	25
+	–	–	<i>a</i>	32	43	29
–	+	–	<i>b</i>	35	34	50
+	+	–	<i>ab</i>	55	47	46
–	–	+	<i>c</i>	44	45	38
+	–	+	<i>ac</i>	40	37	36
–	+	+	<i>bc</i>	60	50	54
+	+	+	<i>abc</i>	39	41	47

Table 1: Treatment Combinations and Results

Exercise 2.

The effect estimates from a 2^4 factorial experiment are listed here. Let say there is 2 replicates in this experiment. Can we decide which factors are significant for $\alpha = 0.05$ level? If yes, write a regression model using only significant factors.

$$ABCD = -2.5251,$$

$$BCD = 4.4054, ACD = -0.4932, ABD = -5.0842, ABC = -5.7696,$$

$$CD = 4.6707, BD = -4.6620, BC = -0.7982, AD = -1.6564, AC = 1.1109, AB = -10.5229,$$

$$D = -6.0275, C = -8.2045, B = -6.5304, A = -0.7914$$

Exercise 3.

Is it possible to have a 2_{III}^{4-1} design? A 2_{IV}^{4-1} design? If yes, give an example. If no, explain why?

Exercise 4.

The researchers use 2^{5-2} design to investigate the effect of A condensation temperature, B amount of material 1, C solvent volume, D condensation time, and E amount of material 2 on yield. The results obtained are as follows:

$$e = 23.2, ad = 16.9, cd = 23.8, bde = 16.8$$

$$ab = 15.5, bc = 16.2, ace = 23.4, abcde = 18.1$$

- (a) Verify that the design generators used were $I = ACE$ and $I = BDE$.
- (b) Write down the complete defining relation and the aliases for this design.
- (c) Estimate the main effects.